University of Utah Fall 2025

MATH 3220-002 PSet 1 Specification

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Subject to Change; Last Updated: 2025-08-24 08:02:54-06:00

1 Background

In MATH 3220 we'll use linear algebra extensively. More accurately, we'll use linear algebra *and geometry*. This first problem set focuses on the more algebraic aspects of linearity, while in the next problem set we'll focus more on the geometric aspects. The problems are designed to build familiarity with linear spaces and transformations from the axiomatic point of view, and reduce matters to matrices and matrix multiplication only when needed.

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The hyperlinks too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version.

This process, part of "perturbing the statement", will enhance your mathematical error detection and correction skills.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. For instance last part of problem 2 below is a paraphrase of Exercise 13 of §8.4 of the book; which in turn merely asks for a proof of Theorem 8.4.6.. Going forward, and possibly for the rest of this problem set, such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let $\mathcal{P}$ be the vector space of polynomials of degree at most 2, with one indeterminate and real coefficients. Consider further the vector

$$p(t) = 3 + t - 6t^2 \in \mathcal{P}.$$

   (a) Let $\mu_1 = 1, \mu_2 = t, \mu_3 = t^2 \in \mathcal{P}$. Then $\mu = (\mu_1, \mu_2, \mu_3)$ is a basis of $\mathcal{P}$.

   (b) Let $\beta_1 = 1 - t^2, \beta_2 = t - t^2, \beta_3 = 2 - 2t + t^2 \in \mathcal{P}$. Then $\beta = (\beta_1, \beta_2, \beta_3)$ is a basis of $\mathcal{P}$.

   (c) Compute the matrix representation $\mathcal{M}_\mu(p)$ of the vector p relative to the basis μ. **There is nothing to prove in this part of the problem!**

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(d) Compute the matrix representation $\mathcal{M}_\beta(\mathrm{p})$ of the vector p relative to the basis β. **There is nothing to prove in this part of the problem!**

(e) Solve

$$\mathrm{A}\mathcal{M}_\beta(\mathrm{p}) = \mathcal{M}_\mu(\mathrm{p})$$

for $\mathrm{A} \in \mathcal{M}(3 \times 3; \mathbb{R})$. **There is nothing to prove in this part of the problem!**

2. Let V, W and X be vector spaces.

(a) For any $\mathrm{L} \in \mathcal{L}(\mathrm{V}; \mathrm{W})$ and for any $\mathrm{K} \in \mathcal{L}(\mathrm{W}; \mathrm{X})$, one has that $\mathrm{K} \circ \mathrm{L} \in \mathcal{L}(\mathrm{V}; \mathrm{X})$. Here $\circ$ is function composition.

(b) For any $\mathrm{L}_0 \in \mathcal{L}(\mathrm{V}; \mathrm{W})$, the function $\mathcal{L}(\mathrm{W}; \mathrm{X}) \to \mathcal{L}(\mathrm{V}; \mathrm{X})$ defined by

$$\mathrm{K} \mapsto \mathrm{K} \circ \mathrm{L}_0$$

is linear.

(c) For any $\mathrm{K}_0 \in \mathcal{L}(\mathrm{W}; \mathrm{X})$, the function $\mathcal{L}(\mathrm{V}; \mathrm{W}) \to \mathcal{L}(\mathrm{V}; \mathrm{X})$ defined by

$$\mathrm{L} \mapsto \mathrm{K}_0 \circ \mathrm{L}$$

is linear.

(d) Suppose additionally that V is n-dimensional with basis $\beta = (\beta_1, \beta_2, ..., \beta_\mathrm{n}) \in \mathrm{V}^\mathrm{n}$, and W is m-dimensional with basis $\gamma = (\gamma_1, \gamma_2, ..., \gamma_\mathrm{m}) \in \mathrm{W}^\mathrm{m}$. For any $\mathrm{L} \in \mathcal{L}(\mathrm{V}; \mathrm{W})$, define $\mathcal{M}_{\gamma \leftarrow \beta}(\mathrm{L})$ to be the $\mathrm{m} \times \mathrm{n}$ matrix defined so that for any $\mathrm{j} \in \{1, 2, ..., \mathrm{n}\}$,

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$$L(\beta_j) = \sum_{i=1}^{m} a_{ij}\gamma_i,$$

where a_{ij} is the ij-th entry of $\mathcal{M}_{\gamma\leftarrow\beta}(L)$. Then

$$\mathcal{M}_{\gamma\leftarrow\beta} : \mathcal{L}(V; W) \to \mathcal{M}(m \times n; \mathbb{R})$$

is a vector space isomorphism.

(e) Suppose additionally that X is p-dimensional vector space with basis $\delta = (\delta_1, \delta_2, ..., \delta_p) \in X^p$. Then for any $L \in \mathcal{L}(V; W)$ and for any $K \in \mathcal{L}(W; X)$,

$$\mathcal{M}_{\delta\leftarrow\gamma}(K)\mathcal{M}_{\gamma\leftarrow\beta}(L) = \mathcal{M}_{\delta\leftarrow\beta}(K \circ L).$$

3. Let X and Y be sets. For a function $f : X \to Y$, the graph of f is by definition the following subset of $X \times Y$:

$$\text{graph}(f) = \{(x, y) \in X \times Y | y = f(x)\}.$$

Let $f : V \to W$ be an arbitrary function between vector spaces. Then f is linear iff its graph is a linear subspace of $V \times W$.

4. Consider the vector space $\mathcal{M} = \mathcal{M}(n\times n; \mathbb{R})$ of $n\times n$ matrices with real entries. The trace tr(A) of a matrix $A \in \mathcal{M}$ is by definition the sum of the diagonal entries of A, that is,

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii},$$

where a_{ij} is the ij-th entry of A.

(a) $\text{tr} : \mathcal{M} \to \mathbb{R}$ is a linear transformation.

(b) The following function is a vector space isomorphism:

$$\Phi : \mathcal{M} \to \mathcal{L}(\mathcal{M};\mathbb{R}), \quad A \mapsto [X \mapsto \mathrm{tr}(AX)].$$

5. Consider the vector space $\mathcal{M} = \mathcal{M}(2 \times 2;\mathbb{R})$ of 2×2 matrices with real entries.

   (a) There are unique functions $\delta, \tau : \mathcal{M} \to \mathbb{R}$ such that for any $A \in \mathcal{M}$,

   $$A^2 = \tau(A)A - \delta(A)I.$$

   (b) For any $A \in \mathcal{M}$ and for any nonnegative integer n, the n-th power A^n of A is in the span of A and I. **It is likely that for this part induction will be useful!**

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a generative AI tool or a computer algebra system for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or

- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., ChatGPT) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the Wayback Machine (see their "Save Page Now" documentation) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as prompt engineering, and it is your responsibility you

design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

– https://platform.openai.com/docs/guides/prompt-engineering
– https://developers.google.com/machine-learning/resources/prompt-eng
– https://www.ibm.com/topics/prompt-engineering
– https://aws.amazon.com/what-is/prompt-engineering/

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

  https://forms.gle/kyqNNDYHBrertPZZA.

  You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

  https://www.gradescope.com/courses/1073533/assignments/6451770, see the Gradescope documentation for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on August 29, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.